

■ Ejercicio 1. Sea

$$f(x, y) = \frac{\sqrt{x^2 - 4}}{\sqrt{xy - x}}$$

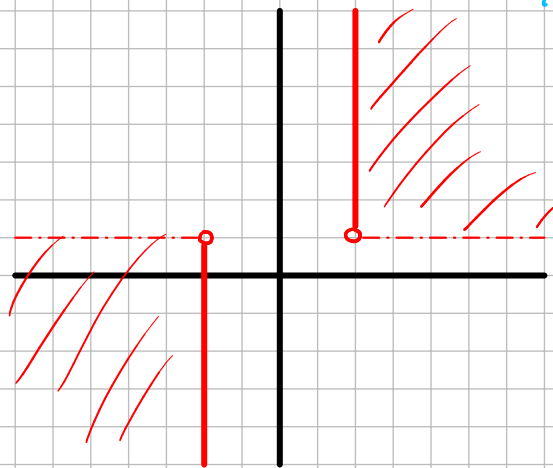
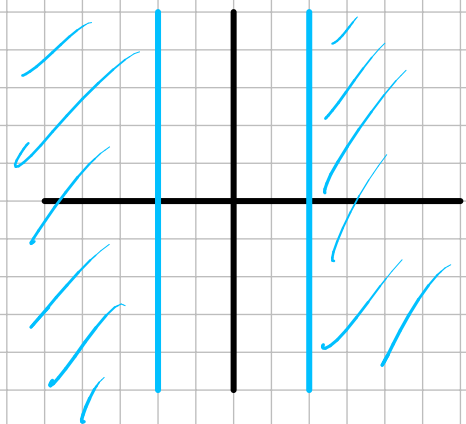
Expresa y grafique el dominio natural de f , indicando si es conexo, abierto y/o cerrado, justificando brevemente su respuesta.

$$xy - x > 0 \quad \wedge \quad x^2 - 4 \geq 0 \rightarrow x^2 \geq 4$$

$$x(y-1) > 0 \quad |x| \geq 2 \rightarrow \begin{matrix} x \geq 2 \\ \vee \\ x \leq -2 \end{matrix}$$

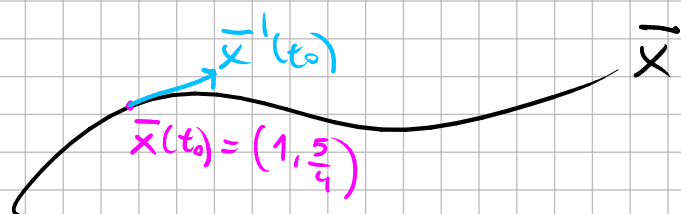
$$x > 0 \wedge y - 1 > 0 \quad \vee \quad x < 0 \wedge y - 1 < 0$$

$$x > 0 \wedge y > 1 \quad \vee \quad x < 0 \wedge y < 1 \quad \wedge \quad x \geq 2 \quad \vee \quad x \leq -2$$



Abierto $\times$
 Cerrado $\times$
 Acotado $\times$
 Compacto $\times$
 Conexo $\times$

- Ejercicio 2. Sean $f(x, y) = ax^2 + xy$ y C la curva de ecuación vectorial $\vec{X} = (2t, t^2 + 1)$, con $t \in \mathbb{R}$. Halle el valor de $a \in \mathbb{R}$ para que en el punto $(1, \frac{5}{4})$ la curva C y el conjunto de nivel de f tengan la misma recta tangente.



$$\vec{X}' = (2, 2t)$$

$$\vec{X}'\left(\frac{1}{2}\right) = (2, 1)$$

$$\begin{aligned} 2t_0 &= 1 \\ t_0^2 + 1 &= \frac{5}{4} \end{aligned} \rightarrow t_0 = \frac{1}{2}$$

$$L: \lambda \cdot (2, 1) + \left(1, \frac{5}{4}\right) = (x, y)$$

$$ax^2 + xy = k$$

$$f(x, y) = ax^2 + xy$$

$$\nabla f = (2ax + y, x) \perp f = k$$

$$\nabla f\left(1, \frac{5}{4}\right) = \left(2a + \frac{5}{4}, 1\right)$$

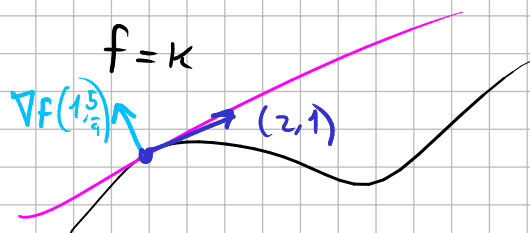
$$\nabla f\left(1, \frac{5}{4}\right) \cdot (2, 1) = 0$$

$$\left(2a + \frac{5}{4}, 1\right) \cdot (2, 1) = 0$$

$$4a + \frac{5}{2} + 1 = 0$$

$$4a = -\frac{7}{2}$$

$$a = -\frac{7}{8}$$



- Ejercicio 3. Sean $h(x, y) = f(\vec{g}(x, y))$, con $\vec{g}(x, y) = (x^2y, y - x^2)$, $y, x + 2y + z = 3$ la ecuación del plano tangente al gráfico de $f \in C^1(\mathbb{R}^2)$ en el punto $(2, 1, f(2, 1))$. Calcule $h(1.02, 1.97)$ utilizando una aproximación lineal.

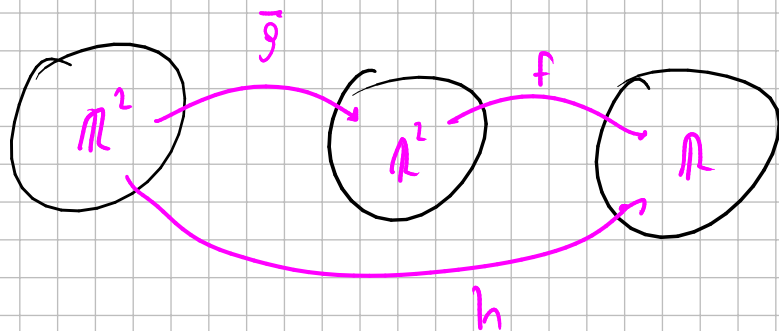
$$h(x, y) \approx h(1, 2) + h'_x(1, 2)(x - 1) + h'_y(1, 2)(y - 2)$$

$$h(1, 2) = f(\vec{g}(1, 2))$$

$$h(1, 2) = f(2, 1)$$

$$z = 3 - x - 2y$$

$$z_0 = 3 - 2 - 2 = -1 = f(2, 1)$$



$$Dh(1, 2) = Df(\vec{g}(1, 2)) \cdot D\vec{g}(1, 2)$$

$$Dh(1, 2) = Df(2, 1) \cdot D\vec{g}(1, 2)$$

$$z = 3 - x - 2y \rightarrow \text{Taylor de 1er orden tomado en } (2, 1)$$

$$z'_x = -1$$

$$z'_y = -2$$

$$Df(2, 1) = [-1 \ -2]$$

$$\vec{g}(x, y) = (x^2y, y - x^2),$$

$$D\vec{g}(x, y) = \begin{bmatrix} 2xy & x^2 \\ -2x & 1 \end{bmatrix}$$

$$D\vec{g}(1, 2) = \begin{bmatrix} 4 & 1 \\ -2 & 1 \end{bmatrix}$$

$$Dh(1,2) = [-1 \ -2] \begin{bmatrix} 4 & 1 \\ -2 & 1 \end{bmatrix}$$

$$Dh(1,2) = [0 \ -3]$$

$$h(x,y) \approx -1 - 3(y-2)$$

$$h(1.02, 1.97) \approx -1 - 3(1.97 - 2)$$

$$h(1.02, 1.97) \approx -0.91$$

■ Ejercicio 4. Sea

$$f(x,y) = x^3 + y^3 - 3xy.$$

Halle los puntos críticos de f y clasifíquelos.

$$\nabla f = (3x^2 - 3y, 3y^2 - 3x) = (0, 0)$$

$$3x^2 - 3y = 0 \quad \wedge \quad 3y^2 - 3x = 0$$

$$x^2 - y = 0 \quad \wedge \quad y^2 - x = 0$$

$$x^2 = y \quad \xrightarrow{\text{pink arrow}} \quad x^4 - x = 0$$

$$x(x^3 - 1) = 0$$

$$x = 0$$

$$x^3 = 1$$

$$x = 1$$

$$y = 0 \quad \leftarrow \text{red arrow from } x=0$$

$$y = 1 \quad \leftarrow \text{red arrow from } x=1$$

$$(0,0) \quad (1,1)$$

$$\nabla f = (3x^2 - 3y, 3y^2 - 3x)$$

$$(0,0) \quad (1,1)$$

P.C.

$$f''_{xx} = 6x$$

$$f''_{xy} = -3$$

$$f''_{yx} = -3$$

$$f''_{yy} = 6y$$

$$Hf = \begin{bmatrix} 6x & -3 \\ -3 & 6y \end{bmatrix}$$

$$Hf(0,0) = \begin{bmatrix} 0 & -3 \\ -3 & 0 \end{bmatrix}$$

$$H_1 = 0$$

No es extremo.

$$H_2 = -9$$

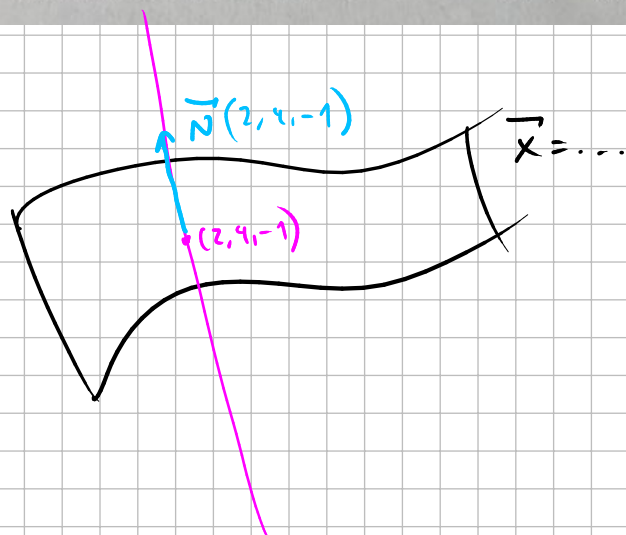
$$Hf(1,1) = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$H_1 = 6$$

$$H_2 = 27$$

MIN local

- Ejercicio 5. Sea Σ la superficie de ecuación $\vec{X} = (uv, 2u^2 + v, u - v)$ con $(u, v) \in \mathbb{R}^2$. Halle la intersección entre la recta normal a Σ en $(2, 4, -1)$ y el plano xy .



$$\vec{X}'_u = (v, 4u, 1)$$

$$\vec{X}'_v = (u, 1, -1)$$

$$uv = 2$$

$$2u^2 + v = 4$$

$$u - v = -1$$

$$v = u + 1$$

$$u(u+1) = 2$$

$$u^2 + u - 2 = 0$$

$$u = 1$$

$$u = -2$$

$$u = 1 \rightarrow v = 2$$

$$2 + 2 = 4 \quad \checkmark$$

$$1 - 2 = -1 \quad \checkmark$$

$$u = -2 \Rightarrow v = -1$$

$$2 \cdot (-2)^2 - 1 = 4$$

$$-1 = -1 \quad \text{Abs.}$$

$$\vec{X}'_u = (v, 4u, 1) = (2, 4, 1)$$

$$\vec{X}'_v = (u, 1, -1) = (1, 1, -1)$$

$$\underline{(-5, 3, -2) = \vec{N}}$$

$$\lambda \cdot (-5, 3, -2) + (2, 4, -1)$$

$$\lambda \cdot (-5, 3, -2) + (2, 4, -1) = (x, y, z)$$

$$(-5\lambda + 2, 3\lambda + 4, -2\lambda - 1) = (x, y, 0)$$

$$-2\lambda - 1 = 0$$

$$\lambda = -\frac{1}{2}$$

$$\Rightarrow x = \frac{5}{2} + 2 = \frac{9}{2}$$

$$y = -\frac{3}{2} + 4 = \frac{5}{2}$$

$$\Rightarrow \left(\frac{9}{2}, \frac{5}{2}, 0\right)$$